LAB 12
Computer Store Management App

Mục tiêu
Xây dựng ứng dụng mobile bằng **React Native (Expo)** để quản lý cửa hàng máy tính, bao gồm:
- CRUD dữ liệu
- Lưu trữ cục bộ bằng SQLite
- Tìm kiếm, lọc, sắp xếp
- Thống kê dữ liệu kinh doanh

Yêu cầu chung
- **Framework:** Expo (React Native)
- **Database:** SQLite
- **Navigation:** Bottom Tab Navigator
- **UI/UX:**
  - Giao diện rõ ràng
  - Responsive cơ bản
  - Trải nghiệm người dùng hợp lý

Data Structure
Mỗi sản phẩm máy tính có cấu trúc:

Field	Type	Description
id	integer	ID tự tăng
product_name	text	Tên sản phẩm
brand	text	Hãng
category	text	Laptop / Desktop / Component
cpu	text	CPU
ram	integer	RAM (GB)
storage	integer	Ổ cứng (GB)
price	real	Giá
stock	integer	Tồn kho
sold_quantity	integer	Đã bán
is_featured	integer	Nổi bật

Navigation Structure
Ứng dụng gồm 4 tab:
1. **Store** – Danh sách sản phẩm
2. **Add Computer** – Thêm sản phẩm
3. **Statistics** – Thống kê
4. **Featured** – Sản phẩm nổi bật

TASK DETAILS

Task 1: Display Product List (Store Tab)
Yêu cầu:
- Sử dụng `FlatList`
- Hiển thị:
  - Product Name
  - Brand
  - Category
  - CPU
  - RAM
  - Storage

    - Price
    - Stock
    - Sold Quantity
    - Icon 🔥 nếu featured

Chức năng:

- Pull-to-refresh
- Click item → Product Detail

Task 2: Add New Computer

Form gồm:

- Product Name
- Brand
- Category (Dropdown)
- CPU
- RAM
- Storage
- Price
- Stock
- Sold Quantity
- Is Featured (Switch)

Validation:

- Không được để trống
- Price > 0
- RAM: 4 → 128
- Storage: 128 → 4096

Sau khi Save:

- Lưu vào SQLite
- Hiển thị thông báo
- Quay về Store

Task 3: Product Detail + Actions

Hiển thị:

- Tất cả thông tin sản phẩm

⚠️ Tính toán:

Sử dụng các công thức sau:

Revenue:

Revenue=price×sold_quantityRevenue = price \times sold_quantityRevenue=price×sold_quantity

Inventory Value:

Inventory Value=price×stockInventory\ Value = price \times stockInventory Value=price×stock

Total Units:

Total Units=stock+sold_quantityTotal\ Units = stock + sold_quantityTotal Units=stock+sold_quantity

Chức năng:

✏️ *Edit*

- Pre-fill form
- Update SQLite

❌ *Delete*

- Confirm dialog:

"Are you sure you want to delete this product?"
- Xóa khỏi DB

Task 4: Search, Filter & Sort

4.1 Search
- Theo Product Name (real-time)

4.2 Filter
- Brand
- Category:
  - Laptop
  - Desktop
  - Component
- Featured (toggle)

4.3 Stock Status
- In Stock (stock > 0)
- Low Stock (stock < 5)
- Out of Stock (stock = 0)

4.4 Advanced Filters
- RAM:
  - 8GB
  - 16GB
  - 32GB+
- Price:
  - < 500
  - 500–1000
  - 1000

4.5 Sort (Bắt buộc)
- Price ↑ / ↓
- Sold Quantity ↑ / ↓

4.6 Reset Filters
- Reset toàn bộ

Task 5: Statistics Tab

Hiển thị:

Tổng sản phẩm
- Total Products

Tổng doanh thu
- SUM(price × sold_quantity)

Tổng đã bán
- SUM(sold_quantity)

Theo Category
- Laptop
- Desktop
- Component

➕ Nâng cao:

Giá trung bình
- AVG(price)

Best Selling Product
- Sản phẩm bán nhiều nhất

Low Stock List
- Danh sách sản phẩm stock < 5

Top Brand

- Brand có tổng sold_quantity cao nhất

Task 6: Featured Tab

Hiển thị:

- Danh sách sản phẩm featured

Thông tin:

- Product Name
- Brand
- Price
- Sold Quantity

Featured Statistics

- Total Featured
- Featured Revenue
- Total Sold (Featured)
- Theo Category

Sample Data (Gợi ý)

Product	Brand	Category	CPU	RAM	Storage	Price	Stock	Sold	Featured
MacBook Pro M3	Apple	Laptop	M3	16	512	1800	10	30	1
Dell XPS 13	Dell	Laptop	i7	16	512	1500	8	20	1
HP Pavilion	HP	Desktop	i5	8	256	800	5	15	0
RTX 4060	Asus	Component	GPU	0	0	400	3	50	1

BONUS (Optional)

- ⭐ Dark Mode
- ⭐ Product Image
- ⭐ Chart (Pie chart Category)

Quick Icon Cheat Sheet - Computer Store Manager

Import

```
import { Ionicons } from '@expo/vector-icons';
```

Usage

```
<Ionicons name="icon-name" size={24} color="color" />
```

Icons by Screen

StoreScreen (Product List)

Icon	Name	Size	Purpose
	laptop-outline	80	Empty state
	add	28	FAB add product
	pricetag-outline	20	Price
	cube-outline	20	Stock
	bar-chart-outline	20	Sold quantity
	flame	20	Featured product
	hardware-chip-outline	20	CPU
	save-outline	20	Storage
	flash-outline	20	Performance highlight
	chevron-forward	18	Navigate to detail

AddComputerScreen

Icon	Name	Size	Purpose
	laptop-outline	64	Header
	create-outline	20	Input field
	hardware-chip-outline	20	CPU field
	save-outline	20	Storage field
	settings-outline	20	Category
	checkmark-circle	20	Save
	close-circle	20	Cancel

ProductDetailScreen

Icon	Name	Size	Purpose
	desktop-outline	64	Header
	hardware-chip	20	CPU
	server-outline	20	Storage
	stats-chart-outline	20	Revenue
	cube	20	Inventory
	add-circle-outline	20	Total units
	create-outline	20	Edit
	trash-outline	20	Delete
	arrow-back-outline	20	Back

Filter & Search UI

Icon	Name	Size	Purpose
	search-outline	20	Search bar
	pricetags-outline	20	Filter brand
	folder-outline	20	Category
	flash	20	Featured filter

	`trending-down-outline`	20	Low stock
	`trending-up-outline`	20	High sales
	`refresh-outline`	20	Reset filters
	`swap-vertical-outline`	20	Sort

StatisticsScreen

Icon	Name	Size	Purpose
	`analytics-outline`	48	Header
	`cube-outline`	24	Total products
	`cash-outline`	24	Revenue
	`bar-chart`	24	Total sold
	`calculator-outline`	24	Average price
	`trophy-outline`	24	Best selling
	`alert-circle-outline`	24	Low stock
	`business-outline`	24	Top brand

FeaturedScreen

Icon	Name	Size	Purpose
	`flame`	48	Header
	`star`	20	Featured badge
	`cash`	20	Revenue
	`stats-chart`	20	Sold
	`grid-outline`	20	Category breakdown

Navigation (Bottom Tabs)

Icon	Name	Size	Purpose
	`home` / `home-outline`	var	Store
	`add-circle` / `add-circle-outline`	var	Add
	`analytics` / `analytics-outline`	var	Statistics
	`flame` / `flame-outline`	var	Featured

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Size Guide

Size	Use Case
16	Small labels
18	Navigation arrows
20	Buttons, fields
24	List items
28	FAB
48	Section headers
64	Screen headers
80	Empty states

🎯 Common Patterns

Icon Button

```
<TouchableOpacity onPress={handleSave}>
  <Ionicons name="checkmark-circle" size={24} color="#4CAF50" />
</TouchableOpacity>
```

Icon with Text (Product Info)

```
<View style={{ flexDirection: 'row', alignItems: 'center' }}>
  <Ionicons name="hardware-chip-outline" size={20} color="#6200EE" />
  <Text style={{ marginLeft: 8 }}>Intel i7</Text>
</View>
```

Dynamic Featured Icon

```
<Ionicons
  name={item.is_featured ? 'flame' : 'flame-outline'}
  size={20}
  color={item.is_featured ? '#FF5722' : '#999'}
/>
```

Stock Status Icon

```
<Ionicons
  name={
    item.stock === 0
      ? 'close-circle'
      : item.stock < 5
      ? 'alert-circle'
      : 'checkmark-circle'
  }
  size={20}
  color={
    item.stock === 0
      ? '#B00020'
      : item.stock < 5
      ? '#FF9800'
      : '#4CAF50'
  }
/>
```

🌈 Suggested Colors

```
colors.primary = '#6200EE'
colors.success = '#4CAF50'
colors.warning = '#FF9800'
colors.error = '#B00020'
colors.textSecondary = '#757575'
colors.featured = '#FF5722'
colors.gold = '#FFC107'
```

🧠 Icon Naming Convention

```
// Active (filled)
<Ionicons name="flame" />

// Inactive (outline)
<Ionicons name="flame-outline" />
```

👉 Dùng:

- **Filled** → active / highlight
- **Outline** → normal state

🔍 Suggested Icon Categories

💻 Computer & Tech

- `laptop`, `desktop`, `hardware-chip`, `server`

Data & Analytics

- `analytics`, `bar-chart`, `stats-chart`, `calculator`

Business

- `cash`, `card`, `wallet`, `pricetag`

Actions

- `add`, `create`, `save`, `trash`, `refresh`

Inventory

- `cube`, `archive`, `layers`

Quick Tips

1. Luôn dùng icon + text

👉 Tránh UI chỉ có icon

2. Consistency

👉 CPU luôn dùng `hardware-chip`

3. Highlight logic

- Featured → flame
- ⚠ Low stock → alert
- Revenue → cash

4. UX nâng cao

- Hover (TouchableOpacity opacity)
- Active tab → icon filled

Tổng kết

Icons quan trọng nhất:

- `laptop-outline` → sản phẩm
- `hardware-chip-outline` → CPU
- `cash-outline` → giá/tiền
- `cube-outline` → kho
- `flame` → featured
- `analytics-outline` → thống kê